

Design Document

TruLabel

UI, Database, Modules, and Algorithm/Flowchart

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1. UI Design

The interface is designed for a non-technical consumer, so the primary flow uses plain-language output rather than exposing raw agent reasoning. Three core screens are planned:

1.1 Search Screen

- A single search bar accepting a product name or brand.
- An optional “upload label photo” action (supports FR-2, OCR-based extraction), shown as a secondary, less prominent option.
- Recent search history displayed below the search bar for quick re-access (supports FR-9).

1.2 Verdict Report Screen

- Prominent display of the overall Marketing Accuracy Score (e.g., a score out of 100, with a simple color band — red/amber/green).
- A per-claim breakdown: each extracted claim shown alongside its verdict (supported / contradicted / unsupported) and a short plain-language explanation.
- An expandable “view evidence” section per claim, showing the specific source and data used — supports NFR-6 (Transparency).
- A persistent disclaimer banner clarifying the verdict is advisory, not a legal/regulatory finding (supports FR-10).
- Action buttons: “Compare with another product” and “Find a better alternative,” entry points into Compare Mode and Recommend Mode respectively.

1.3 Compare / Recommend Screen

- Compare Mode: a side-by-side card layout, one column per product, aligning claims and MAS scores for direct comparison.
- Recommend Mode: the originally searched product shown alongside the system's top recommended alternative, with a short justification (e.g., “Scores higher because its sugar claim is accurate”).

2. Database Design

TruLabel uses a relational schema (PostgreSQL) since claims, evidence, and verdicts have clear, structured relationships to one another. The simplified schema below reflects the core entities and their relationships:

Figure 1: Simplified database schema (entities and relationships)

2.1 Core Tables

Table	Key Fields	Purpose
Product	product_id (PK), name, brand, category, OFF_reference_id	Stores the identified product and its link to its Open Food Facts record.
Claim	claim_id (PK), product_id (FK), claim_text, claim_type	Stores each discrete claim extracted from the product's marketing text.
Evidence	evidence_id (PK), claim_id (FK), source, credibility_score	Stores each piece of evidence retrieved for a claim, with its

		credibility weighting.
Verdict	verdict_id (PK), claim_id (FK), result, explanation	Stores the per-claim verdict (supported / contradicted / unsupported) and its explanation.
Report	report_id (PK), product_id (FK), MAS_score, created_at	Stores the overall Marketing Accuracy Score for a completed verification run.
User Session	session_id (PK), search_history	Tracks a user's search history for FR-9 (non-authenticated, session-scoped).
Comparison / Recommendation	comp_id (PK), product_ids (list), recommended_product_id	Stores the result of a Compare Mode or Recommend Mode run (FR-11–FR-14).

3. Module Design

The system is organised into independently testable modules, aligned with NFR-4 (Maintainability):

Module	Responsibility
Frontend (React)	Search UI, verdict report display, compare/recommend views.
API Gateway / Orchestrator (FastAPI)	Receives requests, invokes the agent pipeline in sequence, returns structured responses.
Claim Extraction Agent	Extracts discrete, checkable claims from marketing text.
Evidence Retrieval Agent	Queries Open Food Facts for ingredient/nutrition ground truth.
Reality-Check Agent	Compares each claim against retrieved evidence and the curated regulatory rule set.
Credibility-Weighting Agent	Assigns a credibility weight to each evidence source used.
Verdict Synthesis Agent	Combines per-claim verdicts into the overall Marketing Accuracy Score and report.
Comparison Agent	Aligns verdicts and evidence across multiple products (FR-11).
Category Candidate Search + Quick Heuristic Filter	Retrieves and cheaply ranks same-category alternatives (FR-12, FR-13).
Recommendation Agent	Selects and justifies the best alternative from the verified shortlist (FR-14).
Database Layer (PostgreSQL)	Persists products, claims, evidence, verdicts, reports, and comparison/recommendation results.

4. Algorithm / Flowchart

4.1 Core Verification Algorithm

The core verification process (used within both single-product verification and, per-product, within Compare and Recommend Modes) follows this sequence:

- Step 1: Accept product search query or label input from the user.
- Step 2: Claim Extraction Agent parses marketing text and outputs a list of discrete claims.
- Step 3: Evidence Retrieval Agent queries Open Food Facts for the product's ingredient/nutrition data.
- Step 4: For each claim, the Reality-Check Agent compares the claim against retrieved data and the curated regulatory rule set, producing a preliminary verdict.
- Step 5: The Credibility-Weighting Agent assigns a weight to each piece of evidence used in Step 4, based on source reliability (regulatory > peer-reviewed/ingredient data > brand-stated).

- Step 6: The Verdict Synthesis Agent aggregates weighted, per-claim verdicts into a single Marketing Accuracy Score (MAS) and generates a plain-language report.
- Step 7: The report is returned to the user with full evidence citations.

Figure 2: Core verification pipeline flowchart

4.2 Extended Modes Algorithm

- Compare Mode: repeat the core verification algorithm (Steps 1–7) independently for each selected product, then invoke the Comparison Agent to align all resulting reports side by side.
- Recommend Mode: (a) retrieve all products in the same Open Food Facts category as the selected product; (b) apply the Quick Heuristic Filter — a non-LLM ranking using raw nutrition/ingredient data (e.g., lower sugar content, fewer additives, better available Nutri-Score) — to shortlist the top few candidates; (c) run the core verification algorithm (Steps 1–7) only on the shortlisted candidates; (d) invoke the Recommendation Agent to select and justify the best-verified alternative.

Figure 3: Extended modes (Compare & Recommend) flowchart

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